DSC5001: Statistical Machine Learning I

Tutorial 4: Linear Regression

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Semester A, 2026/27

Tutorial Overview

This tutorial practices four aspects of linear regression:

1. Derive the ordinary least-squares solution.
2. Quantify uncertainty in regression coefficients.
3. Diagnose regression problems.
4. Implement linear regression in Python.

Tasks 1–3 are required.
Task 4 is optional.

Task 1: Matrix Form of Linear Regression

Goal. Use the matrix form of OLS to compute coefficients and verify the residual property. Consider:

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \qquad y = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$$

The OLS estimator is:

$$\widehat{\beta} = (X^T X)^{-1} X^T y$$

Tasks:

1. Compute $\widehat{\beta}$.
2. Compute fitted values: $\widehat{y} = X\widehat{\beta}$.
3. Verify the residual property:

$$X^T e = 0, \qquad e = y - \widehat{y}.$$

Solution: Task 1

First compute:

$$X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

and:

$$X^T y = \begin{bmatrix} 10 \\ 23 \end{bmatrix}.$$

Therefore:

$$\widehat{\beta} = (X^T X)^{-1} X^T y$$

gives:

$$\boxed{\widehat{\beta} = \begin{bmatrix} \frac{1}{3} \\ \frac{3}{2} \end{bmatrix}}$$

Solution: Task 1 (continued)

Hence the fitted model is:

$$\boxed{\widehat{y} = \frac{1}{3} + \frac{3}{2}x}$$

Using:

$$\widehat{y} = X\widehat{\beta}$$

we obtain:

$$\boxed{\widehat{y} = \begin{bmatrix} \frac{11}{6} \\ \frac{10}{3} \\ \frac{29}{6} \end{bmatrix}}$$

Solution: Task 1 (continued)

The residual vector is:

$$e = y - \widehat{y}$$

so:

$$e = \begin{bmatrix} \frac{1}{6} \\ -\frac{1}{3} \\ \frac{1}{6} \end{bmatrix}.$$

Now verify:

$$X^T e = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{6} \\ -\frac{1}{3} \\ \frac{1}{6} \end{bmatrix} = \boxed{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}.$$

OLS residuals are orthogonal to the columns of the design matrix.

Task 2: Regression Uncertainty

Goal. Understand how the design matrix affects coefficient uncertainty.
Consider a simple regression with:

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}.$$

Assume:$\sigma^2 = 4$.
The covariance of OLS is:

$$\mathrm{Var}(\widehat{\beta} \mid X) = \sigma^2 (X^T X)^{-1}.$$

Tasks:

1. Compute $(X^T X)^{-1}$.
2. Compute $\mathrm{Var}(\widehat{\beta} \mid X)$.
3. Find the standard error of the slope coefficient.
4. Explain how the uncertainty changes if σ^2 increases.

Solution: Task 2

First compute:

$$X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}.$$

Its inverse is:

$$(X^T X)^{-1} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}.$$

Therefore:

$$\boxed{(X^T X)^{-1} = \begin{bmatrix} 7/3 & -1 \\ -1 & 1/2 \end{bmatrix}}$$

Solution: Task 2 (continued)

Using:

$$\mathrm{Var}(\widehat{\beta} \mid X) = \sigma^2 (X^T X)^{-1}$$

and:

$$\sigma^2 = 4,$$

we obtain:

$$\mathrm{Var}(\widehat{\beta} \mid X) = 4 \begin{bmatrix} 7/3 & -1 \\ -1 & 1/2 \end{bmatrix}.$$

Therefore:

$$\boxed{\mathrm{Var}(\widehat{\beta} \mid X) = \begin{bmatrix} 28/3 & -4 \\ -4 & 2 \end{bmatrix}}$$

Solution: Task 2 (continued)

The slope variance is the second diagonal element:

$$\mathrm{Var}(\widehat{\beta}_1) = 2.$$

Hence:

$$\boxed{SE(\widehat{\beta}_1) = \sqrt{2}}$$

Solution: Task 2 (continued)

The coefficient covariance is:

$$\mathrm{Var}(\widehat{\beta} \mid X) = \sigma^2 (X^T X)^{-1}.$$

If:

$$\sigma^2 \uparrow,$$

then coefficient variances increase proportionally.
Therefore:

$$SE(\widehat{\beta}_j) \uparrow,$$

and confidence intervals become wider.

Coefficient uncertainty depends on both the noise level and the information contained in the predictors.

Task 3: Regression Diagnostics

Goal. Use residuals, leverage, and Cook's distance to identify influential observations.
Consider: $x = (1, 2, 3, 4, 10), y = (2, 3, 5, 4, 15)$ with fitted values: $\widehat{y} = (2, 3, 4, 5, 8)$.
Use:

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2},$$

and compare Cook's distance using:

$$D_i \propto e_i^2 \frac{h_{ii}}{(1 - h_{ii})^2}.$$

Tasks:

1. Compute residuals.
2. Compute leverage values.
3. Compare the Cook's distance scores.
4. Identify the most influential observation.

Solution: Task 3 – Residuals

Residuals are:

$$e_i = y_i - \widehat{y}_i.$$

Therefore:

$$\boxed{e = (0, 0, 1, -1, 7)}$$

The largest absolute residual occurs at observation 5:

$$\boxed{|e_5| = 7}$$

Solution: Task 3 – Leverage

For simple regression with an intercept:

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}.$$

For:

$$x = (1, 2, 3, 4, 10),$$

we have:

$$\bar{x} = \frac{1 + 2 + 3 + 4 + 10}{5} = \boxed{4}.$$

Also:

$$\sum_j (x_j - \bar{x})^2 = 9 + 4 + 1 + 0 + 36 = \boxed{50}.$$

Solution: Task 3 – Leverage Values

Since $n = 5$, $\frac{1}{n} = 0.2$.
Therefore:

$$h_1 = 0.2 + \frac{9}{50} = \boxed{0.38},$$

$$h_2 = 0.2 + \frac{4}{50} = \boxed{0.28},$$

$$h_3 = 0.2 + \frac{1}{50} = \boxed{0.22}.$$

Also:

$$h_4 = 0.2 + \frac{0}{50} = \boxed{0.20},$$

and:

$$h_5 = 0.2 + \frac{36}{50} = \boxed{0.92}.$$

Solution: Task 3 – Leverage Interpretation

The leverage vector is therefore:

$$\boxed{h = (0.38, 0.28, 0.22, 0.20, 0.92)}$$

Observation 5 has by far the largest leverage:

$$\boxed{h_5 = 0.92}$$

This occurs because:

$$x_5 = 10$$

is far from the sample mean:

$$\bar{x} = 4.$$

Leverage measures how unusual an observation is in predictor space.

Solution: Task 3 – Cook's Distance

Use the common-factor score:

$$C_i = e_i^2 \frac{h_{ii}}{(1 - h_{ii})^2}.$$

Because $e_1 = e_2 = 0$,

$$\boxed{C_1 = C_2 = 0}.$$

For observation 3:

$$C_3 = 1^2 \frac{0.22}{(1 - 0.22)^2} = \frac{0.22}{0.78^2} \approx \boxed{0.362}.$$

For observation 4:

$$C_4 = (-1)^2 \frac{0.20}{(1 - 0.20)^2} = \frac{0.20}{0.80^2} = \boxed{0.313}.$$

Solution: Task 3 – Cook's Distance

For observation 5:

$$C_5 = 7^2 \frac{0.92}{(1-0.92)^2}.$$

Thus:

$$C_5 = 49 \frac{0.92}{0.08^2} = \boxed{7043.75}.$$

Therefore the relative Cook's-distance scores are approximately:

$$\boxed{C = (0,\ 0,\ 0.362,\ 0.313,\ 7043.75)}.$$

These scores differ from the exact Cook's distances only by the common positive scaling factor omitted in the proportional formula.

Solution: Task 3 – Influence

Observation 5 has:

$$\boxed{|e_5| = 7}$$

and:

$$\boxed{h_5 = 0.92}.$$

It therefore combines:

$$\boxed{\text{large residual} + \text{very high leverage}}.$$

Its Cook's-distance score is also overwhelmingly larger than the others. Therefore:

$$\boxed{\text{Observation 5 is the most influential observation.}}$$

Optional Task 4: Regression Modeling and Diagnostics

Goal. Build a regression pipeline and evaluate both prediction performance and model assumptions.
Use the California Housing dataset from scikit-learn.
Tasks:

1. Split the data into training and test sets.
2. Fit a mean-response baseline and a linear regression model.
3. Compare test MSE.
4. Plot residuals against fitted values and inspect the pattern.
5. Report the coefficient with the largest absolute value and discuss its interpretation.

Solution: Task 4 – Data Preparation

Load the required packages and dataset:

```
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split

data = fetch_california_housing()

X = data.data
y = data.target
```

Solution: Task 4 – Train/Test Split

Split training and test data:

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=0
)
```

The test set is kept unseen during model fitting.

Solution: Task 4 – Mean Baseline

The baseline predicts the training-set mean for every test observation:

```
baseline = np.mean(y_train)

y_pred_base = np.ones(
    len(y_test)
) * baseline
```

This provides a simple reference prediction.

Solution: Task 4 – Linear Regression

Fit linear regression:

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()

model.fit(
    X_train,
    y_train
)

y_pred = model.predict(X_test)
```

Solution: Task 4 – Test MSE

Compute test MSE:

```
from sklearn.metrics import mean_squared_error

mse_base = mean_squared_error(
    y_test,
    y_pred_base
)

mse_linear = mean_squared_error(
    y_test,
    y_pred
)

print(mse_base)
print(mse_linear)
```

A smaller test MSE indicates better predictive performance on the held-out data.

Solution: Task 4 – Residual Diagnostics

Compute and plot residuals:

```
residual = y_test - y_pred

plt.scatter(
    y_pred,
    residual,
    s=12
)

plt.axhline(0)
plt.xlabel("Fitted values")
plt.ylabel("Residuals")
plt.show()
```

Look for systematic curvature, changing spread, or unusual observations.

Solution: Task 4 – Coefficients

Inspect the fitted coefficients:

```
feature_names = data.feature_names
coef = model.coef_

idx = np.argmax(np.abs(coef))

print(feature_names[idx])
print(coef[idx])
```

The sign describes the fitted direction of association while holding the other predictors fixed.

Common mistake

The predictors have different units, so raw coefficient magnitudes should not be used directly to rank feature importance.

Solution: Task 4 – Interpretation

A complete analysis should discuss:

- whether linear regression improves over the mean baseline;
- whether residuals show systematic structure;
- whether unusual observations are present;
- what the fitted coefficients mean in their original units.

$$\boxed{\text{prediction performance} + \text{diagnostic evidence}}$$

Both are needed to evaluate the fitted regression model.

Tutorial 4: Linear Regression

Questions and discussion